

UNIVERSITY INSTITUTE OF ENGINEERING

COMPUTER SCIENCE & ENGINEERING

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OPERATING SYSTEM(24CST-205/24ITT-205)

Unit No.3

Chapter No. 3

Lecture No.37

Topic : Access Matrix and Implementation of Access Matrix

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Learning Objectives

- Understand the concept of the access matrix as a model for defining access rights between subjects (users/processes) and objects (resources).
- Analyze how access control is represented using rows (domains) and columns (objects).

Access Matrix

- View protection as a matrix (access matrix)
- Rows represent domains
- Columns represent objects
- $\text{Access}(i, j)$ is the set of operations that a process executing in Domain_i can invoke on Object_j

Access Matrix

object domain	F_1	F_2	F_3	printer
D_1	read		read	
D_2				print
D_3		read	execute	
D_4	read write		read write	

Use of Access Matrix

- If a process in Domain D_i tries to do “op” on object O_j , then “op” must be in the access matrix
- User who creates object can define access column for that object
- Can be expanded to dynamic protection
 - Operations to add, delete access rights
 - Special access rights:
- owner of O_i

Use of Access Matrix

copy op from O_i to O_j (denoted by “*”)

control – D_i can modify D_j access rights

transfer – switch from domain D_i to D_j

- Copy and Owner applicable to an object
- Control applicable to domain object

Use of Access Matrix (Cont.)

- Access matrix design separates mechanism from policy
- Mechanism
 - Operating system provides access-matrix + rules
 - If ensures that the matrix is only manipulated by authorized agents and that rules are strictly enforced
- Policy
 - User dictates policy
 - Who can access what object and in what mode
- But doesn't solve the general confinement problem

Access Matrix of Figure A with Domains as Objects

domain \ object	F_1	F_2	F_3	laser printer	D_1	D_2	D_3	D_4
D_1	read		read			switch		
D_2				print			switch	switch
D_3		read	execute					
D_4	read write		read write		switch			

Access Matrix with Copy Rights

domain \ object	F_1	F_2	F_3
D_1	execute		write*
D_2	execute	read*	execute
D_3	execute		

(a)

domain \ object	F_1	F_2	F_3
D_1	execute		write*
D_2	execute	read*	execute
D_3	execute	read	

(b)

Access Matrix With Owner Rights

object domain	F_1	F_2	F_3
D_1	owner execute		write
D_2		read* owner	read* owner write
D_3	execute		

(a)

object domain	F_1	F_2	F_3
D_1	owner execute		write
D_2		owner read* write*	read* owner write
D_3		write	write

(b)

Modified Access Matrix of Figure B

object domain	F_1	F_2	F_3	laser printer	D_1	D_2	D_3	D_4
D_1	read		read			switch		
D_2				print			switch	switch control
D_3		read	execute					
D_4	write		write		switch			

Implementation of Access Matrix

- Generally, a sparse matrix
- Option 1 – Global table
 - Store ordered triples $\langle domain, object, rights-set \rangle$ in table
 - A requested operation M on object O_j within domain $D_i \rightarrow$ search table for $\langle D_i, O_j, R_k \rangle$
4 with $M \in R_k$
 - But table could be large $\rightarrow$ won't fit in main memory
 - Difficult to group objects (consider an object that all domains can read)
- Option 2 – Access lists for objects
 - Each column implemented as an access list for one object
 - Resulting per-object list consists of ordered pairs $\langle domain, rights-set \rangle$ defining all domains with non-empty set of access rights for the object
 - Easily extended to contain default set $\rightarrow$ If $M \in$ default set, also allow access

Implementation of Access Matrix (Cont.)

- Option 3 – Capability list for domains
 - Instead of object-based, list is domain based
 - **Capability list** for domain is list of objects together with operations allowed on them
 - Object represented by its name or address, called a **capability**
 - Execute operation M on object O_j , process requests operation and specifies capability as parameter
 - 4 Possession of capability means access is allowed
 - Capability list associated with domain but never directly accessible by domain
 - 4 Rather, protected object, maintained by OS and accessed indirectly
 - 4 Like a “secure pointer”
 - 4 Idea can be extended up to applications
- Option 4 – Lock-key
 - Compromise between access lists and capability lists
 - Each object has list of unique bit patterns, called **locks**
 - Each domain as list of unique bit patterns called **keys**
 - Process in a domain can only access object if domain has key that matches one of the locks



Comparison of Implementations

- Many trade-offs to consider
 - Global table is simple, but can be large
 - Access lists correspond to needs of users
 - 4 Determining set of access rights for domain non-localized so difficult
 - 4 Every access to an object must be checked
 - Many objects and access rights -> slow
 - Capability lists useful for localizing information for a given process
 - 4 But revocation capabilities can be inefficient
 - Lock-key effective and flexible, keys can be passed freely from domain to domain, easy revocation
- Most systems use combination of access lists and capabilities
 - First access to an object -> access list searched
 - 4 If allowed, capability created and attached to process
 - Additional accesses need not be checked
 - 4 After last access, capability destroyed
 - 4 Consider file system with ACLs per file

Summary of the Lecture

This lecture enables the students to understand the concept of access matrix is a protection model that defines the access rights of users or processes over system resources. It organizes permissions in a structured way to enforce security and controlled resource sharing.



Next Lecture

- **The Security Problem**

Quiz/ FAQ's

Q1. In an access matrix, rows represent:

- a) Objects
- b) Domains (subjects/users)
- c) Memory blocks
- d) Files only

Q2. The columns in an access matrix correspond to:

- a) Users
- b) Processes
- c) Objects (resources)
- d) Permissions only

References

<https://www.includehelp.com/c-programming-questions/>

<https://www.studytonight.com/operating-system/>

<https://computing.llnl.gov/tutorials/>

[https://www.tutorialspoint.com/operating_system/index.htm#:~:text=An%20operating%20system%20\(OS\)%20is,software%20in%20a%20computer%20system.](https://www.tutorialspoint.com/operating_system/index.htm#:~:text=An%20operating%20system%20(OS)%20is,software%20in%20a%20computer%20system.)

<https://www.javatpoint.com/os-tutorial>

<https://www.guru99.com/operating-system-tutorial.html>

<https://www.geeksforgeeks.org/operating-systems/>



Faculty-curated videos, NPTEL, Coursera, LinkedIn, or other relevant learning resources

- <https://nptel.ac.in/courses/106105214>
- <https://www.linkedin.com/learning/iot-foundations-operating-systems-fundamentals-2018>



Class-Wise Feedback

Faculty members are encouraged to generate a class specific QR code at the end of each session to collect brief student feedback. This can support self-reflection, improve teaching practices, and allow students to share insights, suggestions, or concerns anonymously.



Thank You

For queries

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